

Pinning Effects of Heavy Elements for the Structural Stability of Ni-Based Layered Oxides

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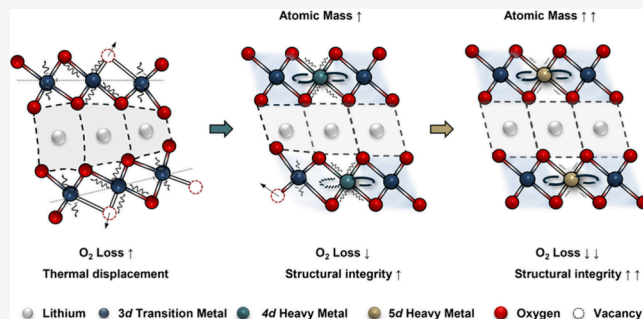


Article Recommendations



Supporting Information

ABSTRACT: Ultrahigh-Ni layered oxides enable high-energy lithium-ion batteries (LIBs) but are plagued by lattice collapse, O release, and rapid capacity fade under high temperature and voltage operation. Here, 0.25 mol % Nb and Ta are incorporated into $\text{LiNi}_{0.92}\text{Co}_{0.04}\text{Mn}_{0.04}\text{O}_2$ to elucidate the effect of mass on structural stability. Despite identical oxidation states (+5) and radii (0.64 Å), their mass difference reveals distinct pinning effects. X-ray diffraction and absorption spectroscopy analyses exhibit reduced atomic displacement, reinforced Ni–O bond strength, and expanded Li slab spacing, improving electrochemical performance upon various operating conditions. Moreover, thermal analysis confirms suppressed O release from the lattice structure and a delayed decomposition reaction, with more pronounced stabilization from heavier elements. These findings underscore an atomic-mass-driven materials design strategy as an effective approach for enhancing the durability of ultrahigh-Ni layered cathodes in next-generation LIBs.



As the demand for high-energy-density lithium-ion batteries (LIBs) continues to grow, layered transition metal oxides, particularly ultrahigh-Ni $\text{LiNi}_x\text{M}_{1-x}\text{O}_2$ ($x \geq 0.9$, $M = \text{Co, Mn, Al, etc.}$), have gained significant attention as cathode materials due to their high specific capacity and relatively low cost compared to Co-rich alternatives.^{1–14} These issues become particularly pronounced at high voltages, where the cathode undergoes complex phase transitions, transition metal migration, and O evolution, all of which contribute to lattice collapse and mechanical stress.

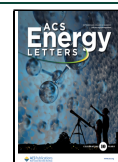
One of the most critical structural instabilities in Ni-rich layered oxides is the H2-to-H3 phase transition, which occurs at high states of charge (SoC). This transition is associated with repulsive interactions between adjacent O layers upon Li^+ extraction.^{15–17} At elevated voltages, this phase transition induces a pronounced lattice contraction along the c -axis of the layered structure, significantly impacting the material's structural integrity and electrochemical performance.¹⁸ The abrupt lattice contraction induces internal mechanical strain,¹⁹ leading to the formation of microcracks,^{20–22} particle pulverization,^{23,24} and a loss of electrical connectivity, all of which accelerate capacity degradation and voltage decay. Furthermore, at elevated temperatures, the layered structure of

NCM materials becomes thermodynamically unstable, undergoing a sequential phase transformation.¹¹ During this process, transition metal cations migrate from the transition metal layer into Li vacancies, exacerbating cation disorder and promoting O release from the lattice.^{25,26} This not only deteriorates the structural integrity of the cathode but also poses severe safety risks due to the exothermic nature of the O evolution and subsequent electrolyte decomposition. To address these challenges, various strategies have been proposed, including elemental doping, surface coating, etc.^{27–31} Among these approaches, elemental doping has emerged as one of the most effective methods for stabilizing the Ni-rich cathode structure. By incorporating dopant elements into the crystal lattice, doping enhances structural integrity through lattice stabilization, suppression of transition metal migration, reduction of O

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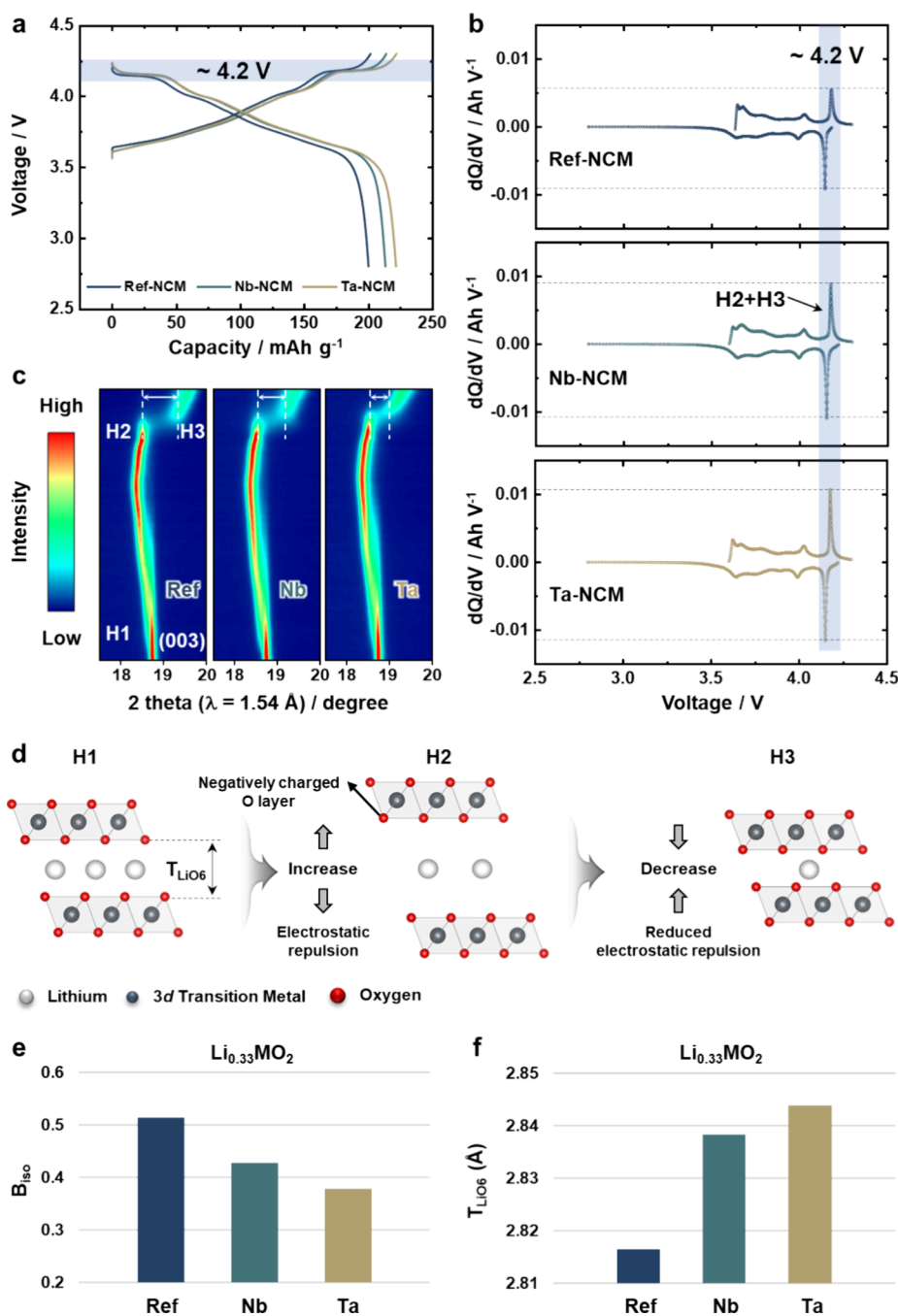


Figure 1. (a) Charge–discharge voltage profiles of Ref-NCM, Nb-NCM, and Ta-NCM. (b) dQ/dV curves of each material. (c) Contour plots of *in situ* XRD patterns recorded during the charge process for Ref-NCM, Nb-NCM, and Ta-NCM, and the corresponding (d) schematic presentation of H1, H2, and H3. (e) Calculated B_{iso} and (f) T_{LiO_6} with 0.33 mol of Li^+ remaining in the crystal structure (~ 4.2 V).

evolution, and mitigation of interfacial side reactions. Elements such as Mg,^{32–34} Al,^{35–37} Ti,^{38–40} Zn,^{41,42} Zr,^{43–46} Nb,^{47–49} Mo,^{50,51} Ta,^{52–55} and W^{55–59} have been extensively investigated for their ability to reinforce the cathode framework, thereby improving electrochemical performance and long-term stability.

In addition, although it is not specifically focused on Ni-rich materials, there is a report that interprets the enhancement of structural stability in layered materials through heavy element doping from a mass-based perspective. Zhao et al.⁶⁰ investigated Sn doping in LiCoO_2 and demonstrated that Sn incorporation suppresses lattice vibrations, stabilizes the CoO_6

octahedral framework, and prevents the irreversible transition to the cubic spinel phase under high voltage cycling conditions. Further analysis based on Debye theory suggested that heavy metal dopants can effectively suppress thermal lattice vibrations, provided that the parent structure's bonding characteristics remain largely unaltered. These findings highlight the potential of heavy element doping as a universal approach to enhancing the stability of layered cathodes.

Building upon this concept, we systematically investigate the effects of incorporating Nb and Ta into a $\text{Li-Ni}_{0.92}\text{Co}_{0.04}\text{Mn}_{0.04}\text{O}_2$ (NCM) ultrahigh-Ni layered cathode. Nb^{5+} and Ta^{5+} exhibit the same oxidation state (5+) and have

identical ionic radii ($0.64 \text{ \AA}$)⁶¹ but differ in atomic weight, making them ideal candidates to examine how heavy metal dopants influence crystal structure, phase transition dynamics, and electrochemical stability. This design enables a systematic investigation of the isolated mass effect, which has been challenging to decouple from ionic size and charge influences in previous doping studies. Additionally, their strong bonding affinity with anions can suppress O evolution, enhance structural rigidity, and stabilize the H3 phase transition, ultimately leading to improved cycle life and high voltage stability. Thus, using a comprehensive set of advanced characterization techniques, including X-ray diffraction (XRD), X-ray absorption spectroscopy (XAS), transmission X-ray microscopy (TXM), and electrochemical analysis, we analyze how Nb and Ta incorporation influences the crystal and electronic structure of the NCM cathode. Through this, we examine the effects of these dopants on electrochemical and thermal decomposition reactions, providing critical insights into the role of heavy metal dopants in suppressing structural degradation and improving thermal stability.

To investigate the effects of incorporating heavy metal elements into ultrahigh-Ni layered oxide materials, 0.25 mol % Nb and Ta were introduced into the NCM material. Electron probe microanalysis (EPMA) mapping was conducted to confirm the presence and distribution of the dopant elements within the particles (Figure S1). Line scan analysis along a predefined horizontal axis revealed that Nb⁵⁺ and Ta⁵⁺ signals were absent at the particle boundary but became detectable toward the interior, indicating a uniform distribution of these elements throughout the particle matrix. Detailed characterization of the pristine state crystal and electronic structures, obtained via XRD and XAS, is provided in Figure S2 and Table S1 of the Supporting Information. Notably, the atomic displacement parameter (B_{iso}), primarily governed by thermal lattice vibrations and static disorder,⁶² exhibited a decreasing trend upon Nb and Ta incorporation. Given that static disorder is generally minor in structurally similar materials, the reduction in B_{iso} can be primarily attributed to suppressed thermal lattice vibrations.⁶⁰ The attenuation of lattice vibrations is typically associated with the presence of heavier elements and stronger chemical bonds within the structure. In this study, the decrease in B_{iso} upon Nb and Ta doping suggests that the introduction of these heavy elements enhances structural stability by mitigating thermal lattice vibrations. This is further supported by extended X-ray absorption fine structure (EXAFS) analysis, which revealed a reduction in the Debye–Waller factor and an increase in Ni–O bond strength in the order from Ref-NCM to Nb-NCM to Ta-NCM (Figure S3 and Table S2).

The electrochemical behaviors of Ref-NCM, Nb-NCM, and Ta-NCM were evaluated to assess the impact of Nb and Ta doping on capacity and structural stability during cycling. The specific capacity was observed to increase in the order of Ref-NCM < Nb-NCM < Ta-NCM over the voltage range of 2.8–4.3 V (Figure 1a). This trend is further supported by the differential capacity (dQ/dV) curves, where an increase in peak intensity at approximately 4.2 V indicates an extended voltage plateau in this region, correlating with a higher capacity (Figure 1b). Such a tendency remains over subsequent cycles (Figure S4). This enhancement is attributed to the more stable progression of the H2-to-H3 phase transition upon Nb and Ta doping. To validate this hypothesis, *in situ* XRD measurements were performed during the charge process, and the resulting

contour plot is presented in Figure 1c. The corresponding quantitative values of the (003) d -spacing changes associated with the H2 and H3 phases are provided in Figure S5. The data reveal that in Ref-NCM, the H2-to-H3 phase transition occurs abruptly, whereas in Ta-NCM, the transition exhibits a more pseudo-one phase behavior. In layered oxide materials, the removal of Li⁺ during charging reduces its initial screening effect, leading to lattice expansion due to increased repulsion between O layers (Figure 1d). At higher SoC, significant Li⁺ depletion increases the oxidation state of the transition metal, enhancing transition metal–O covalency. This reduces the electron density in the O layer, ultimately causing lattice contraction and initiating the H3 phase transition. To further examine the state of the H2-to-H3 transition at high voltages, Rietveld refinement was performed on the crystal structures of the materials at a charged state corresponding to Li = 0.33, where the H2 and H3 phases coexist (Figures S6 and S7 and Table S3). In the charged state, the B_{iso} values exhibit a decreasing trend (Ref-NCM > Nb-NCM > Ta-NCM), while the T_{LiO6} values show an increasing trend (Figure 1e,f), both of which are consistent with those observed in the pristine state. This suggests that the H2-to-H3 phase transition at an equivalent SoC exhibits enhanced structural stability and more favorable Li⁺ transport pathways in the doped materials, following the order Ref-NCM < Nb-NCM < Ta-NCM. As a result, the enhanced phase transition stability in Nb-NCM and Ta-NCM at high voltages contributes to increased capacity while mitigating lattice collapse, leading to improved long-term cycling performance (Figure S8). This effect is further corroborated by the observed decrease in particle pulverization after 100 cycles (Figure S9). Moreover, comprehensive electrochemical property measurements, including electrochemical impedance spectroscopy (EIS) and galvanostatic intermittent titration techniques (GITT) (Figure S10), together with performance evaluations such as rate capability, voltage decay analysis, C/2 long-term cycling, high voltage cycling (4.4 to 2.8 V), and high temperature cycling (45 and 60 °C) (Figure S11), consistently revealed a property and performance hierarchy of Ref-NCM < Nb-NCM < Ta-NCM. In addition to the mass-dependent pinning effect, previous studies have reported that heavy element doping can also influence microstructural development through grain boundary-related mechanisms. For instance, Rathore et al. showed that high-valence, heavy dopants such as W tend to segregate to grain boundaries rather than being uniformly incorporated into the bulk lattice, forming amorphous secondary phases that inhibit primary grain growth, strengthen grain boundaries, and suppress intergranular cracking during cycling.⁶³ Similarly, Qu et al. demonstrated that Ta⁵⁺/Ti⁴⁺ codoping in Ni-rich layered oxides results in refined primary grain sizes and reinforced grain boundaries, accompanied by a mixed-cation surface layer that mitigates H2–H3 phase transitions and reduces crack propagation.⁶⁴ Although our analyses indicate that Nb and Ta are distributed throughout the secondary particles, it is plausible that partial segregation to internal grain boundaries may occur at the nanoscale, contributing to improved resistance to microcracking and enhanced mechanical integrity. Such microstructural strengthening effects, working in concert with the intrinsic mass-driven suppression of Ni–O bond vibrations, could further account for the superior electrochemical performance observed in the Nb- and Ta-doped ultrahigh-Ni layered oxides.

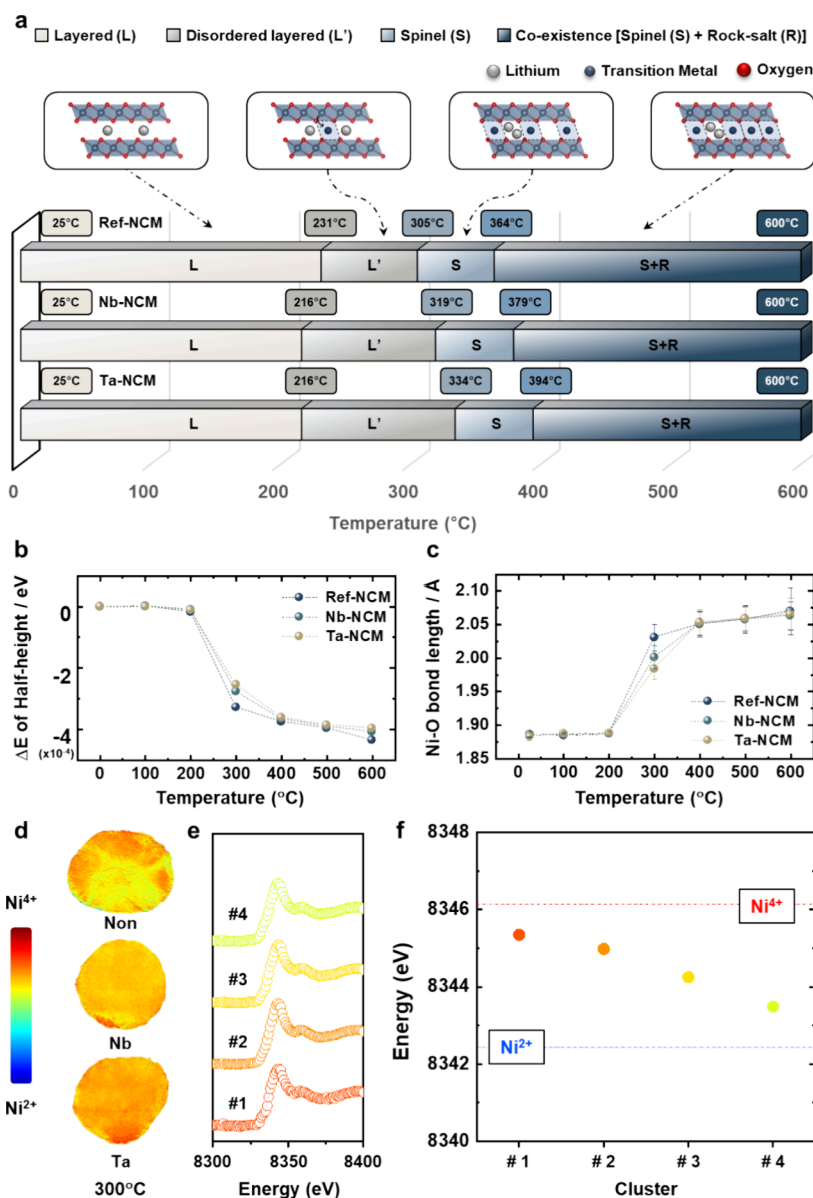


Figure 2. (a) Bar graph illustrating the thermal decomposition sequence reconstructed from *in situ* XRD patterns for charged ($x = 0.33$ in Li_xMO_2) Ref-NCM, Nb-NCM, and Ta-NCM cathode materials, measured at a heating rate of $2.5\text{ }^\circ\text{C}/\text{min}$ from 250 to $600\text{ }^\circ\text{C}$. The phases are denoted as L (layered), L' (disordered layered), S (spinel), and R (rock-salt). (b) Variation of the Ni K-edge half-height position and (c) Ni–O bond length during the thermal decomposition reaction. (d–f) TXM-XANES results measured at room temperature after heating to $300\text{ }^\circ\text{C}$ for charged Ref-NCM, Nb-NCM, and Ta-NCM. (d) Chemical distribution maps for Ni in each particle. (e) Normalized Ni K-edge XANES spectra extracted from four clustered regions of interest (ROIs) and (f) corresponding Ni K-edge half-height energy value.

To evaluate the impact of heavy metal doping on the thermal stability of NCM materials, the phase transition behavior was examined during heating to $600\text{ }^\circ\text{C}$ at each charged state (Figure S12). All three samples exhibited a sequential thermal decomposition pathway: layered (L) $\rightarrow$ disordered layered (L') $\rightarrow$ spinel (S) $\rightarrow$ rock-salt (R) phase (Figure 2a). This thermal decomposition reaction occurs as transition metal ions in the transition metal layer migrate into vacant Li sites due to externally applied thermal energy. As the disorder progresses, a significant number of transition metal ions migrate, leading to the reordering of Li and transition metal species with O evolution, ultimately forming spinel and rock-salt phases. A notable observation is that the onset temperature for the formation of the disordered layered phase is lower in Nb- and Ta-doped samples compared to the Ref-

NCM. However, the temperatures required for the formation of spinel and rock-salt phases increase in the order of Ref-NCM $<$ Nb-NCM $<$ Ta-NCM, indicating enhanced thermal stability with Nb and Ta doping. To further investigate the thermal decomposition mechanism, an XAS experiment was performed at $100\text{ }^\circ\text{C}$ intervals from room temperature to $600\text{ }^\circ\text{C}$. The Ni K-edge X-ray absorption near edge structure (XANES) spectra indicate a progressive shift to lower energy with increasing temperature (Figure S13). In addition, Ni K-edge EXAFS fitting was performed to obtain information about the local structure (Figures S14–S16 and Tables S4–S6). Up to $200\text{ }^\circ\text{C}$, no significant changes were observed in the oxidation state of Ni or in the Ni–O bond length (Figure 2b,c). However, at $300\text{ }^\circ\text{C}$, the oxidation state of Ni decreased rapidly, accompanied by an increase in the Ni–O bond length.

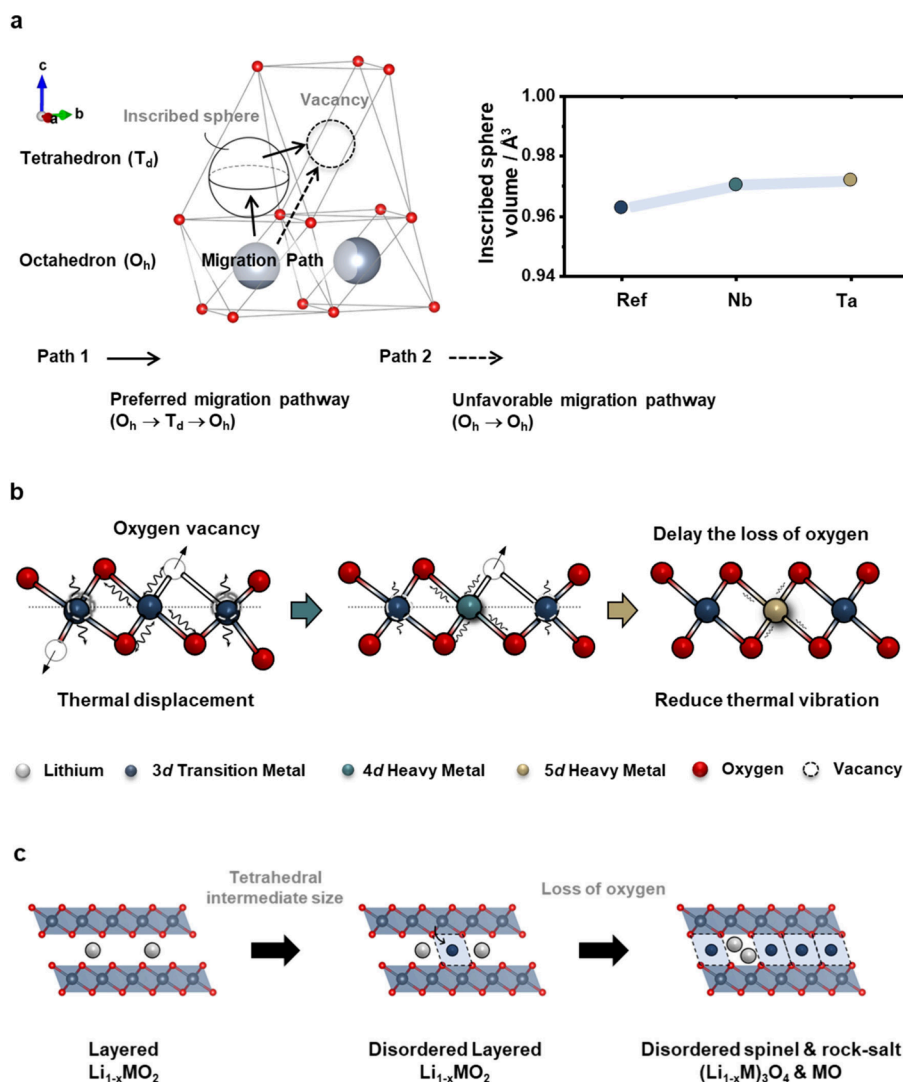


Figure 3. Schematic representation of the cation migration pathway during the thermal decomposition reaction, along with the corresponding inscribed sphere volume within the tetrahedral framework for charged ($\text{Li}_{0.33}\text{MO}_2$) Ref-NCM, Nb-NCM, and Ta-NCM materials. (b) Illustration of the pinning effect induced by heavy elements within the transition metal layer. (c) Simplified schematic depiction of key factors influencing each step of the thermal decomposition reaction.

This suggests that O_2 evolution occurs predominantly between 200 and 300 $^\circ\text{C}$, with the extent of Ni reduction following the trend Ref-NCM > Nb-NCM > Ta-NCM, further confirming the thermal stabilizing effect of Nb and Ta doping. Moreover, using ab initio molecular dynamics (AIMD) simulation, we demonstrated the pinning effect in the Ni–O bonding interaction induced by Nb and Ta doping after delithiation (Figure S17). In the case of the delithiated Ref-NCM, most of the Ni–O bond distances were approximately 1.86 $\text{\AA}$, and the distribution of the bond lengths was also narrow at 0 K. However, it was confirmed that the delithiated Nb-NCM and Ta-NCM exhibited a more diverse distribution of Ni–O bond distances compared to the delithiated Ref-NCM due to the introduction of the heavy metal dopants Nb and Ta. As a result, even when the temperature increases to 500 K, Nb-NCM and Ta-NCM exhibit smaller variations in the Ni–O bond distances compared to Ref-NCM. Furthermore, the temperature-dependent changes in the average Ni–O bonding distance were also smaller in Nb-NCM and Ta-NCM compared to Ref-NCM. These AIMD simulation results support the experimental results, suggesting that Nb-NCM

and Ta-NCM possess higher structural stability than Ref-NCM after delithiation. TXM analysis was conducted to assess localized changes in oxidation states within individual particles at elevated temperatures (Figure 2d).⁶⁵ Ni K-edge XANES spectra were extracted from four distinct clusters within each particle, and their half-height edge energies were plotted (Figure 2e,f). The color gradient from blue to red represents an increasing number of Ni oxidation states. After the sample was heated to 300 $^\circ\text{C}$ and cooled to room temperature, a more pronounced distribution of reduced Ni regions with lower oxidation states was observed in Ref-NCM compared to Nb-NCM and Ta-NCM. This observation aligns with the XAS results, further confirming that Nb and Ta doping mitigates the extent of Ni reduction and enhances structural stability under thermal stress.

The formation of the disordered layered structure begins with the migration of transition metal ions through tetrahedral sites adjacent to the Li layer (Path 1) rather than directly through the shortest path (Path 2) to the empty octahedral sites within the Li layer (Figure 3a). This suggests that the larger the tetrahedral sites, the lower the activation energy

barrier for cation migration, thus decreasing the onset temperature of the phase transition. The right side of Figure 3a presents the calculated volume of the tetrahedral inscribed sphere in the charged state for each material. While Nb- and Ta-doped NCM materials exhibit similar tetrahedral sizes, both have an overall larger size than Ref-NCM. This increase indicates a more energetically favorable environment for cation migration when Nb and Ta are incorporated, which explains the lower phase transition onset temperature observed in the doped materials. Following the initiation of thermal decomposition, characterized by transition metal migration as confirmed by XRD, the subsequent decomposition reaction is progressively delayed in the order of Ref-NCM < Nb-NCM < Ta-NCM. This can be attributed to the strong bonding interactions of Nb–O and Ta–O, as well as the pinning effect of heavy elements, which effectively mitigates oxygen evolution and thermal vibration (Figures 3b,c). As a result, the release of O from the lattice and the associated thermal decomposition reaction are significantly suppressed in Nb- and Ta-doped NCM materials.

Here, we demonstrated that the incorporation of heavy metal elements into an ultrahigh-Ni layered material significantly enhances its structural stability, electrochemical performance, and thermal stability. XRD and XAS analyses confirmed that heavy metal doping led to a decrease in B_{iso} and an increase in Ni–O bond strength, indicating enhanced structural stability. Notably, this effect persisted even in the high voltage region, suggesting improved resistance to lattice distortions and Ni–O bond instability under extreme electrochemical conditions. Consequently, the H2-to-H3 phase transition, a critical factor in high voltage operation, is observed to progress more smoothly in heavy-metal-doped NCM, mitigating abrupt lattice contraction and mechanical stress. Electrochemical measurements confirm that heavy-metal-doped materials exhibit higher specific capacities and an improved cycle life compared to undoped NCM. The increased T_{LiO_6} and reduced B_{iso} suggest that these dopants create a more favorable Li^+ transport environment while suppressing thermal lattice vibrations. *In situ* XRD and dQ/dV analyses further indicate that the H3 phase is stabilized through the pinning effect of Nb and Ta, resulting in a prolonged voltage plateau and enhanced electrochemical stability at high voltages. Thermal stability investigations reveal that while Nb- and Ta-doped NCM exhibits a lower onset temperature for the formation of the disordered layered phase, the subsequent decomposition pathway, involving O_2 gas evolution and transition metal migration, is significantly delayed. This is attributed to the strong Nb–O and Ta–O bonding interactions, as well as the pinning effect of heavy elements, which effectively suppresses O_2 evolution within the lattice, reducing the extent of phase transformation into spinel and rock-salt phases. Notably, these stabilizing effects were more pronounced for Ta-doped NCM, highlighting the role of atomic mass in modulating the structural and thermal stability of the material. Thus, the synergistic effects of heavy element doping, including enhanced structural stability, smoother phase transitions, suppressed lattice collapse, and delayed thermal decomposition, can make these materials promising candidates for high-energy-density LIB cathodes. These findings offer valuable insights into dopant engineering strategies for stabilizing layered oxide materials and extending the operational lifespan of next-generation rechargeable batteries.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenenergylett.5c02563>.

Experimental section; Rietveld refinement data for pristine and $\text{Li}_{0.33}\text{MO}_2$ states; electron probe microanalysis mapping images of Nb/Ta distribution; schematic illustration of oxygen bonding forces; EXAFS spectra and fitting results; dQ/dV curves; (003) *d*-spacing variations; voltage profiles; experimental and refined XRD patterns; capacity retention data; cross-sectional SEM images after cycling; EIS data and Li^+ diffusion coefficients; rate capability and long-term cycling performance; *in situ* XRD patterns upon heating; XANES spectra at various heat-treatment temperatures; and AIMD-simulated structures and radial distribution functions (PDF)

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Notes

The authors declare no competing financial interest.

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